Layer	Tool/ platform/ software	Description
Processor class	CPU	Our standard CPUs in desktops and laptops. It is quite possible to run DL programs in a CPU. Particularly when beginners and intermediate learners get their hands dirty with DL practices, it is quite possible to run programs in laptops and desktops. However, generally available information indicates that real-life or competition (like Kaggle) DL programs tend to run slower in CPUs compared to other processor variants.
	GPU	Graphical processing unit. These were initially meant for gamers. But they've got a new use case with the ubiquitous proliferation of DL programs. They tend to perform better for real scale or competition-oriented DL problems. Companies like Nvidia have various offerings in this space, including US$ 99 Nano cards.
	TPU	Tensor processing unit. This is essentially an ASIC (Application Specific Integrated Circuit) targeted at AI related problems and was developed by Google. The company found most of its use cases around Internet of Things (IoT) Edge devices, sensors and actuators.
Low level language layer	CUDA cuDNN	Compute Unified Device Architecture, that Nvidia is the proprietor of. It enables developers to write DL programs specially optimised and targeted for CUDA enabled GPUs. cuDNN is the CUDA Deep Neural Network library. It has specialised functionality inbuilt as primitives for DL problems, targeted for GPUs. The latest standard of CUDA is almost like C++. So essentially, things are moving towards the openly available language alternatives even for GPU accelerated platforms.
	BLAS	Basic Linear Algebra Subprograms. The reference BLAS is a freely available software package. This is essentially a specification designed for linear algebra and matrix operations. Python's NumPy can also be used for BLAS compatible operations. It even has a Fortran implementation called Eigen.
Library	TensorFlow	This famous ML/DL library is from Google. It was first released in 2015, is open source, and can be run in IoT devices as well as on the mobile. It even has JavaScript support, which means things can be done in the browser as well.
	Theano	This is another open source alternative of TensorFlow, developed by the University of Montreal and has been released under the BSD3 licence. It is Python based and can support CUDA as the underlying platform also.
	CNTK	Cognitive Toolkit. This is another open source library from Microsoft, for almost the same purpose as TensorFlow and Theano.
	PyTorch	This is also an open source library for ML problems, initially developed by Facebook's AI research lab. It is a Python implementation of the Torch library.
High level programming layer	Keras	For beginners as well as experienced practitioners, Keras is the place of go for writing DL programs using plain old Python. It's completely FOSS. One can install and just start writing the very first 'Hello world' sort of DL program within minutes. Another benefit of Keras is that one need not necessarily fiddle with libraries like TensorFlow, Theano or CNTK, nor bother whether the platform is a CPU/GPU. It can be targeted for any platform and any underlying processor. There are good books available on how to learn Keras using Python (2.7 or 3.6). One such book is: 'Deep Learning with Python' by Francois Chollet from Manning.

Table 1: The taxonomy and tools for ML/DL